

LECTURE-15 (BIOFUELS)

➤ INTRODUCTION

- Biofuel, is a fuel (gas, liquid or solid) derived from biomass.
- Generally, gases and liquids fuels obtained from biomass are considered as biofuels.
- The biomass may be plant or algae material or animal waste.
- Since such feedstock material can be replenished readily, biofuel is considered to be a source of renewable energy, unlike fossil fuels such as petroleum, coal, and natural gas.
- Biofuel is commonly advocated as a cost-effective and environmentally benign alternative to petroleum and other fossil fuels, particularly within the context of rising petroleum prices and increased concern over the contributions made by fossil fuels to global warming.
- Many critics express concerns about the scope of the expansion of certain biofuels because of the economic and environmental costs associated with the refining process and the potential removal of vast areas of arable land from food production.
- Crops used to make biofuels are generally either high in sugar (such as sugarcane, sugarbeet, and sweet sorghum), starch (such as maize and tapioca) or oils (such as soybean, rapeseed, coconut, sunflower etc.).
- The source and the biofuel have been depicted by the figures given below:

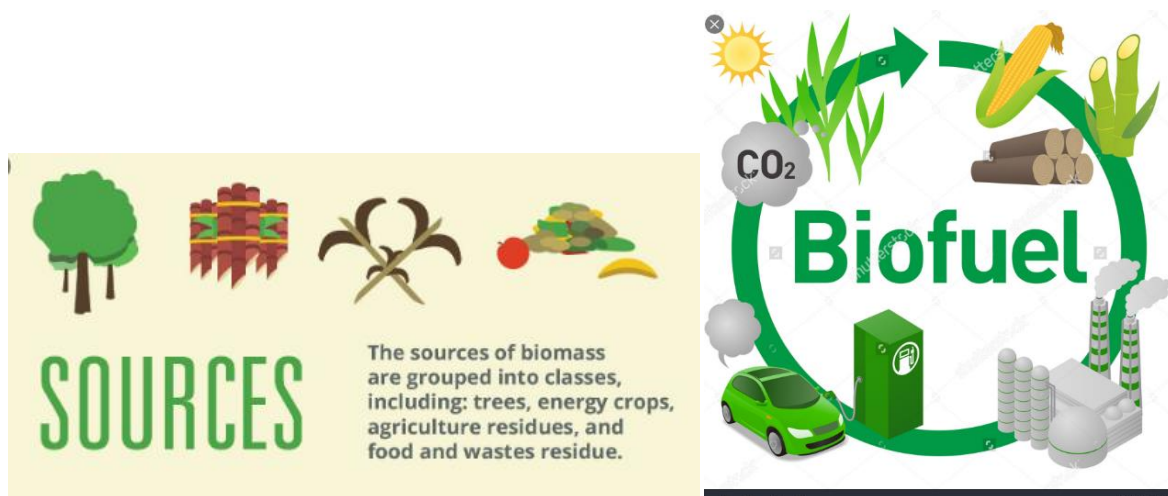


Figure-1: The source and biofuel.

- **Biofuel Types**
 - ✓ The term biofuel refers to liquid or gaseous fuels for the transport sector that are predominantly produced from biomass.
 - ✓ It is generally held that biofuels offer many benefits, including sustainability, reduction of greenhouse gas emissions, and security of supply.
 - ✓ A variety of fuels can be produced from biomass resources including liquid fuels, such as ethanol, methanol, biodiesel, and Fischer-Tropsch diesel, and gaseous fuels, such as hydrogen and methane.

- ✓ Biofuels are primarily used in vehicles but can also be used in engines or fuel cells for electricity generation.
- **Relevance of Biofuel Technology**
 - ✓ There are several reasons why biofuels are considered relevant technologies by both developing and industrialized countries.
 - ✓ They include energy security, environmental concerns, foreign exchange savings, and socioeconomic issues related to the rural sector.
 - ✓ Due to the widespread availability of biomass resources, biomass-based fuel technology can potentially employ more people than fossil fuel-based technology.
 - ✓ Demand for energy is increasing every day due to the rapid growth of population and urbanization.
 - ✓ As the major conventional energy sources like coal, petroleum, and natural gas are gradually depleted, biomass is emerging as one of the promising environmentally friendly renewable energy options.
 - ✓ Due to its environmental merits, the share of biofuel in the automotive fuel market will grow fast in the coming future.
 - ✓ The advantages of biofuels are: (a) they are easily available from common biomass sources, (b) carbon dioxide cycle occurs in combustion, (c) they are very environmentally friendly, and (d) they are biodegradable and contribute to sustainability.
 - ✓ Various scenarios have led to the conclusion that biofuels will be in widespread use in the future energy system.
 - ✓ The scenarios are to facilitate the transition from the hydrocarbon economy to the carbohydrate economy by using biomass to produce bioethanol and biomethanol as replacements for traditional oil-based fuels and feedstocks.
 - ✓ The biofuel scenario produces equivalent rates of growth in GDP and per-capita affluence, reduces fossil energy intensities of GDP, and reduces oil imports.
 - ✓ Each scenario has advantages whether in terms of rates of growth in GDP, reductions in carbon dioxide emissions, the energy ratio of the production process, the direct creation of jobs, or the area of biomass plantation required to make the production system feasible.
- **How Biofuels differ from Petroleum Feedstocks?**
 - ✓ The biggest difference between biofuels and petroleum feedstocks is oxygen content.
 - ✓ Biofuels have oxygen levels of 10 to 45% while petroleum has essentially none, making the chemical properties of biofuels very different from those of petroleum.
 - ✓ All have very low sulfur levels and many have low nitrogen levels.
 - ✓ Biomass can be converted into liquid and gaseous fuels through thermochemical and biological methods.
 - ✓ Biofuel is a non-polluting, locally available, accessible, sustainable, and reliable fuel obtained from renewable sources.
- **Biomass as a Feedstock for Biofuels**
 - ✓ Biomass is an attractive feedstock for three main reasons.
 - ✓ First, it is a renewable resource that could be sustainably developed in the future.

- ✓ Second, it appears to have formidably positive environmental properties resulting in no net releases of carbon dioxide and very low sulfur content.
 - ✓ Third, it appears to have significant economic potential provided that fossil fuel prices increase in the future.
 - ✓ Lignocellulosic biomethanol has such low emissions because the carbon content of the alcohol is primarily derived from carbon that was sequestered in the growing of the biofeedstock and is only being rereleased into the atmosphere.
 - ✓ Carbohydrates (hemicelluloses and cellulose) in plant materials can be converted into sugars by hydrolysis.
 - ✓ Fermentation is an anaerobic biological process in which sugars are converted into alcohol by the action of microorganisms, usually yeast.
 - ✓ The resulting alcohol is bioethanol.
 - ✓ The value of any particular type of biomass as feedstock for fermentation depends on the ease with which it can be converted into sugars.
- **Sources of Liquid Biofuels for Automobiles**
 - ✓ The sources of liquid biofuels are summarized and depicted in the Figure-2, below:

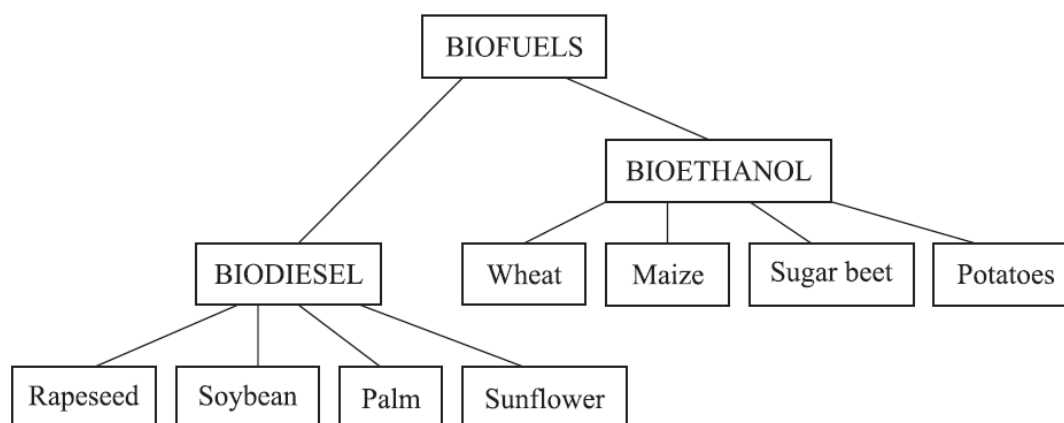


Figure-2: Sources of liquid biofuels for automobiles

➤ **Brief Notes on the Biofuels**

✓ **Bioethanol**

- Bioethanol is ethyl alcohol obtained during fermentation with sugars, starches, or cellulosic biomass.
- It is a petrol additive/substitute.
- It is possible that wood, straw, and even household wastes may be economically converted into bioethanol.
- Ethanol demand is increasing day by day.
- For the supply to be available to meet this demand, new technologies must be moved from the laboratories to commercial reality.
- World ethanol production is about 60% from sugar-crop feedstock. Ethanol is the most widely used liquid biofuel.
- Most commercial production of ethanol is from sugar cane or sugar beet, as starches and cellulosic biomass usually requires expensive pre-treatment.

- Ethanol is used as a renewable energy fuel source as well as for the manufacture of cosmetics and pharmaceuticals and also for the production of alcoholic beverages.
- Ethyl alcohol is not only the oldest synthetic organic chemical used by humans, but it is also one of the most important.

✓ **Biogas**

- Anaerobic digestion of biowastes occurs in the absence of air; the resulting gas, called biogas, is a mixture consisting mainly of methane and carbon dioxide.
- Biogas is a valuable biofuel that is produced in digesters filled with feedstock like dung or sewage.
- The digestion is allowed to continue for a period of ten days to a few weeks.

✓ **The Fischer–Tropsch (FT) Synthesis**

- The **Fischer–Tropsch process** is a collection of chemical reactions that converts a mixture of carbon monoxide and hydrogen into liquid hydrocarbons.
- These reactions occur in the presence of metal catalysts, typically at temperatures of 150–300 °C (302–572 °F) and pressures of one to several tens of atmospheres.
- The process was first developed by Franz Fischer and Hans Tropsch in Germany, in 1925.
- The Fischer-Tropsch (FT) synthesis has been investigated over decades as an alternative route to obtain synthetic fuels from synthesis gas.
- FT is a high-performance synthesis based on metallic catalysis, mainly using ruthenium, cobalt and iron catalysts, which converts syngas in hydrocarbons and chemical precursors.
- The syngas production from biomass gasification becomes the feedstock for subsequent conversion into biofuels through the Fischer-Tropsch synthesis.
- The usage of biomass, includes the lignocellulosic residues, as a raw material in the gasification process.
- Biosyngas is high-lighted as a synthetic fuel source to replace nonrenewable, conventional fossil fuels.
- Lignocellulosic material must be considered a low-cost feedstock to the liquid biofuel production on a large scale.
- Recent understanding of reaction kinetics and thermodynamics has contributed to increasing the FT performance and economic viability.
- Fischer-Tropsch (FT) synthesis, is a crucial reaction for the transformation of non-petroleum carbon resources such as coal, natural gas, shale gas, coal-bed gas and biogas, as well as biomass into liquid fuels and chemicals.
- Temperature, pressure and catalyst behaviour can influence the FT synthesis activity.
- Catalyst factors, including the chemical state of active phases, the promoters, the size and the microenvironment of active phase, determine the CO conversion activity and the product selectivity, particularly the selectivity to C₅₊ hydrocarbons.
- FT synthesis produces hydrocarbons of different lengths from a gas mixture of H₂ and CO (syngas) resulting from biomass gasification.
- The FTS process is capable of producing liquid hydrocarbon fuels from bio-syngas.

- The process for producing liquid fuels from biomass, which integrates biomass gasification with FTS, converts a renewable feedstock into a clean fuel.
- FTS has been explained by the following Figure-3 (a, b):

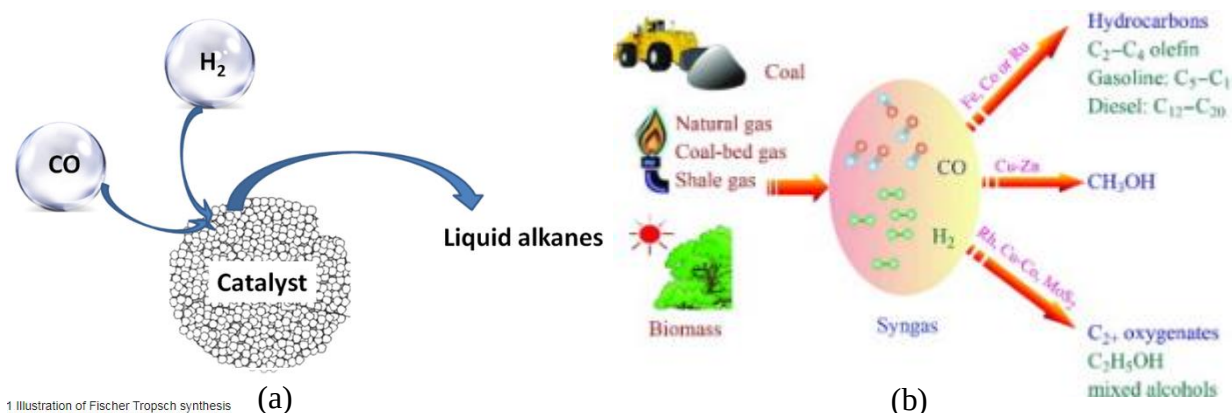


Figure-3: Fischer-Tropsch synthesis for biofuels.

- The products of FTS are mainly aliphatic straight chain hydrocarbons (C_xH_y).
 - Besides the C_xH_y, branched hydrocarbons, unsaturated hydrocarbons, and primary alcohols are also formed in minor quantities.
 - The products obtained from FTS include the light hydrocarbons methane (CH₄), ethene (C₂H₄) and ethane (C₂H₆), LPG (C₃-C₄, propane and butane), gasoline (C₅-C₁₂), diesel fuel (C₁₃-C₂₂), and waxes (C₂₃-C₃₃).
 - The distribution of the products depends on the catalyst and the process parameters such as temperature, pressure, and residence time.
 - The FTS has been extensively investigated and reported in the literature by many researchers .
- ✓ **Vegetable Oils**
- Vegetable oils from renewable oil seeds can be used when mixed with diesel fuels.
 - Pure vegetable oil, however, cannot be used in direct-injection diesel engines, such as those regularly used in standard tractors, since engine cooking occurs after several hours of use.
 - Conversion of vegetable oils and animal fats into biodiesel has been undergoing further development over the past several years.
 - Biodiesel represents an alternative to petroleum-based diesel fuel.
 - Chemically speaking, biodiesel is a mixture of monoalkyl esters of fatty acids, most often obtained from extracted plant oils and/or collected animal fats.
 - Commonly accepted biodiesel raw materials include the oils from soy, canola, corn, rapeseed, and palm.
 - New plant oils that are under consideration include mustard seed, peanut, sunflower, and cotton seed.
 - The most commonly considered animal fats include those derived from poultry, beef, and pork.

➤ GLOBAL SCENARIO W.R.T. BIOFUELS

- In recent years, bioenergy has drawn attention as a sustainable energy source that may cope up with rising energy prices, but also may provide income to poor farmers and rural communities around the globe.
- Rising fuel prices, growing energy demand, concerns over global warming and increased openness to renewable energy resources, domestic energy security and the push for expansion into new markets for crops in the face of world trade outlooks are all factors driving interest in expanding the use of bioenergy.
- Despite keen interest in this sector, there are currently few players in this field.
- Biofuels are fuels that are usually processed from organic matter obtained from living organisms or their products (Biomass).
- They can be used as an alternative to fossil fuels.
- Increasing fuel prices, rising energy demand and global warming issues are the major reasons that drive an enormous interest in exploring natural as well as renewable sources to meet the demand for fuels and energy.
- Over the past 5 years, Biofuels are considered as an alternative to oils on worldwide basis.
- Their reduced carbon emissions in comparison to conventional fuels and their positive impacts on rural development, together with the current high oil prices, are key elements behind their market development.
- Researchers are trying to explore a wide variety of feedstocks, mainly non-edible crops and wastes for generation of cost-effective, high yield and environmental friendly bioenergy that have minimum emissions.
- Worldwide status of bioethanol production in 2017 (country wise) has been shown in Figure-4.

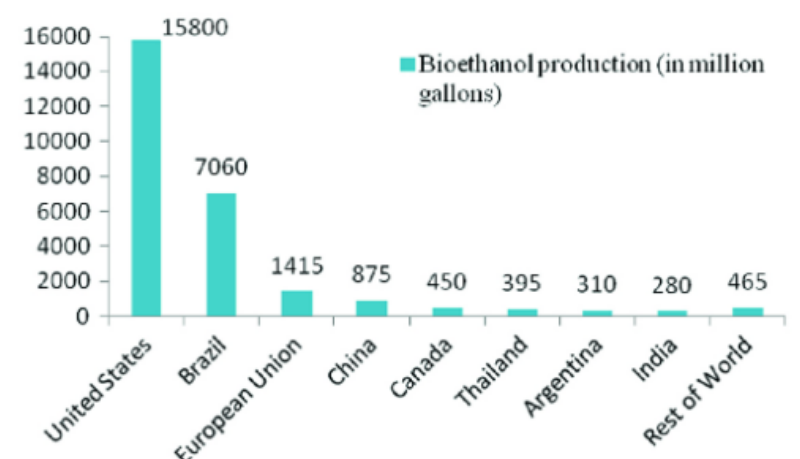


Figure-4: Global status of bioethanol production in 2017.

- United States and Brazil are world leader in the field of bioethanol production.
- However, India being one of the largest importer of energy, has very poor ranking.

- The worldwide projection of biodiesel production in 2017 (countrywise) has been depicted by the Figure-5.

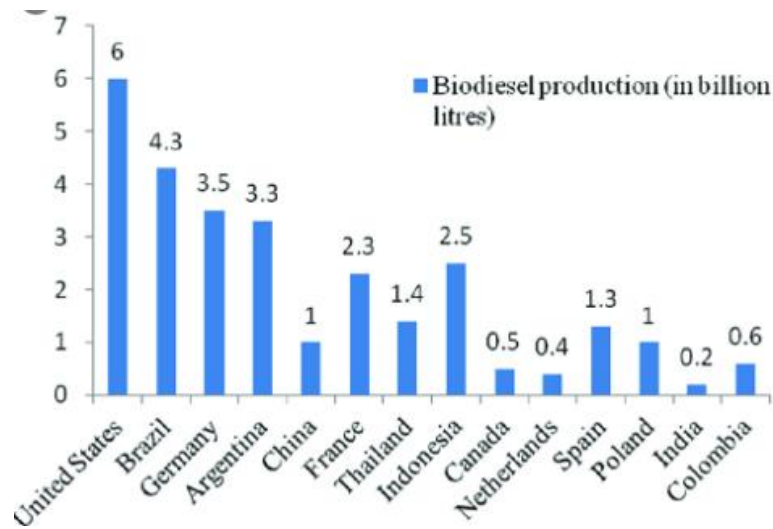


Figure-5: Global status of biodiesel production in 2017.

- Due to the large amount of diesel fuel demands worldwide and the negative environmental and health impacts of its direct combustion, biodiesel production and consumption have been globally increasing as the best short-term substitute for mineral diesel.
- However, using edible and non-edible oil feedstocks for biodiesel production has led to several controversial issues including feedstock availability and cost, greenhouse gas (GHG) emission, land use changes (LUC), and fuel vs. food/feed competition.
- Fortunately, these problems can be effectively overcome using non-crop feedstocks.
- Since 2000, the global biofuels supply has increased by many folds up to the year 2010 as evident from the **Figure-6**. This significant rise is attributed to policies such as blending mandates, which foster greater utilization.

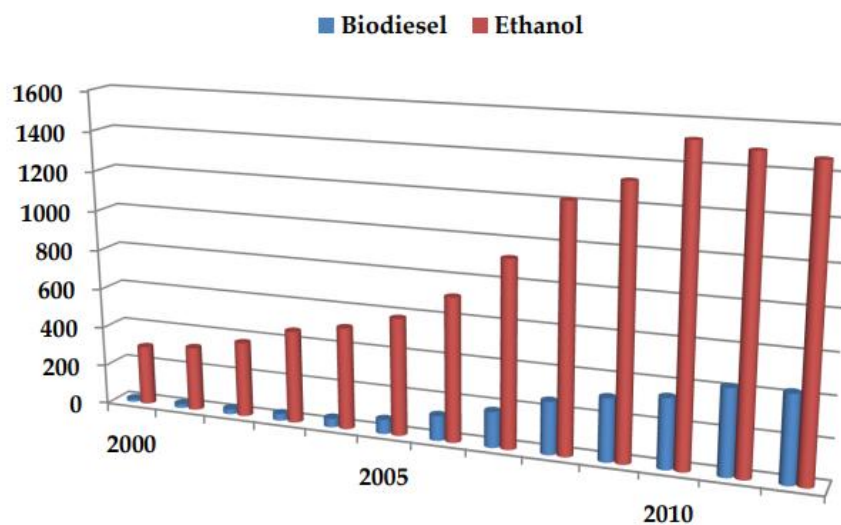


Figure-6: Global biofuel production by fuel type (thousand barrels per day).

- Looking at the broad numbers and types, the global biofuel supply equaled approximately 3:1 breakdown of ethanol to bio-diesel for the year 2013-2015 as clear from the **Figure-7**.
- Conventional biofuels produced from sugar, starch, vegetable oil, or animal fat reflected the majority of the supply.

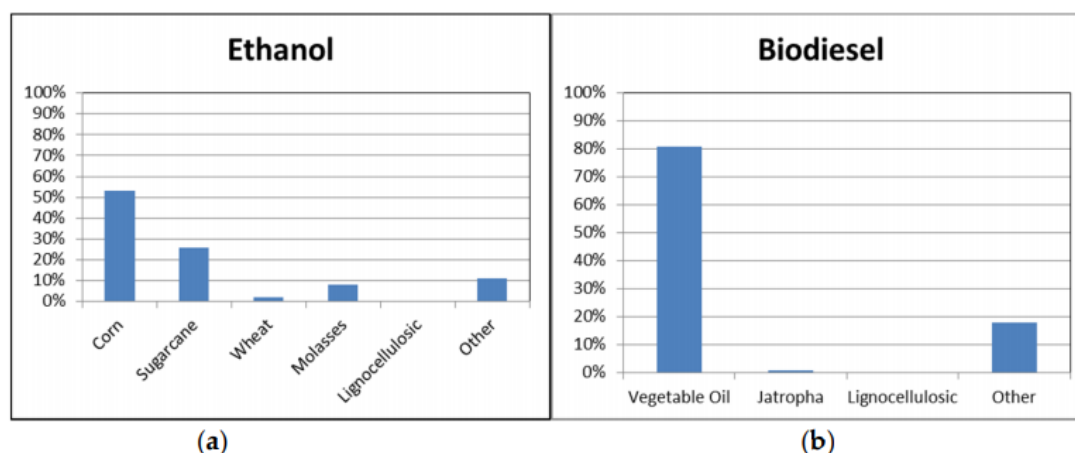


Figure-7: Shares of global ethanol (a) and biodiesel (b) production by feedstock type.

- In recent years, vegetable-based biodiesel nearly matched the ethanol supply produced from sugarcane.

➤ **KEY FEEDSTOCKS FOR BIOFUEL PRODUCTION**

Today about 60% of ethanol is produced from maize, 25% from sugar cane, 7% from molasses, 4% from wheat, and the remainder from other grains, cassava or sugar beets. About 77% of biodiesel is based on vegetable oils (30% soybean oil, 25% pal oil, 18% rapeseed oil) or waste cooking oils (22%). More advanced technologies based on cellulosic feedstocks (e.g. crop residues, wood, or dedicated energy crops) do not account for large shares of total biofuel production. Nevertheless, they are often seen as relevant technologies for the future as they are supposed to cause less competition with food products and emit safer levels of greenhouse gas emissions. The international biofuel sectors are strongly influenced by national policies with three major goals: farmer support, reduced greenhouse gas emissions, and/or reduced energy independency. Table-1 show biofuel production ranking and key feedstocks associated.

Table-1: Biofuel Production Ranking and Key Feedstocks

	Production ranking (base period)		Major feedstocks	
	Ethanol	Biodiesel	Ethanol	Biodiesel
United States	1 (50%)	2 (19%)	Maize	Soybean oil / diverse other oils
European Union	4 (5%)	1 (36%)	Maize / wheat / sugar beet	Rapeseed oil / waste oils
Brazil	2 (24%)	3 (12%)	Sugar cane	Soybean oil
China	3 (8%)	8 (3%)	Maize	Waste oils
India	5 (2%)	15 (0.5%)	Molasses	Palm oil
Canada	6 (1.6%)	10 (1.4%)	Maize	Waste oils
Indonesia	23 (0.2%)	4 (10%)	Molasses	Palm oil
Argentina	9 (1%)	5 (7%)	Maize / sugar cane	Soybean oil
Thailand	7 (1.5%)	6 (4%)	Molasses / cassava	Palm oil
Colombia	13 (0.4%)	9 (1.5%)	Sugar cane	Palm oil
Paraguay	15 (0.3%)	19 (0.03%)	Maize / sugar cane	Soybean oils

Note: Percentage numbers refer to the production share of countries in the base period.

Source: OECD/FAO (2019), “OECD-FAO Agricultural Outlook”, OECD Agriculture statistics (database), <http://dx.doi.org/10.1787/agr-outl-data-en>.

1. Biodiesel includes renewable diesel (also known as Hydrotreated Vegetable Oil or HVO) in the accounting of this *Outlook* although both are different products.

2. An analysis of the potential contribution of biofuels to climate change mitigation in the transport sector (COM/TAD/CA/ENV/EPOC(2018)19/FINAL) that uses the Aglink-Cosimo model finds, however, that a substantially increased role for biofuels in decarbonisation of the transport sector would require a different set of policy incentives that would also need to be cost-effective and take into account effects on food security and the sustainability of resource use.

The diversity of feedstock options for biofuels is fairly significant with local conditions typically framing the choice. The following section and **Table-2** expand on this.

▪ **Lignocellulose**

- ✓ Lignocellulosic material is derived from non-edible crops that have the advantage of limiting cropland expansion and related emissions, with appropriate practices.
- ✓ This plentiful feedstock can be obtained from many different sources, including switchgrass, trees, and agricultural crop residues, such as rice straw, wheat straw, corn stover, and sugarcane bagasse.
- ✓ Depending on the source, there is a large amount of available land.
- ✓ Looking at straw for example, 2.3 billion tons of straw were available in 2011, which has the theoretical potential of making 560 million tons of ethanol.
- ✓ Climate and water needs for lignocellulose vary depending on the source of lignocellulose that is used .
- ✓ There is an incentive to utilize this non-food crop in lieu of traditional corn.
- ✓ The challenge is to produce the fuel economically as the process involves breaking down fibrous plant walls into sugars, which is an expensive step.
- ✓ Once sugars are formed, they can be fermented to produce cellulosic ethanol.

▪ **Algae**

- ✓ Algae refers to a group of photosynthetic organisms, which has promising potential for biofuels in terms of high oil content, limited waste streams and minimal land requirements (compared to biomass), depending on the production pathway.

- ✓ Water is essential for algae cultivation, and can include fresh water, brackish, saline, and wastewater types.
 - ✓ Methods of cultivation and recovery vary with implications for energy and environmental effects.
 - ✓ Data have been limited to date, so there is a fair amount of uncertainty associated with environmental impacts of this feedstock.
 - ✓ As of 2011, current environmental impacts were deemed negligible, as scaled production had not been demonstrated.
- **Corn**
- ✓ Corn (maize) is a fundamental food staple that can be grown in a range of climates from tropical to temperate, and may be sensitive to frost.
 - ✓ Fertilizer and pesticide needs are high for this crop.
 - ✓ For feedstock and ethanol production, water needs are relatively low on the unit basis of ethanol produced.
 - ✓ The United States leads the world in using corn to produce ethanol for fuel.
- **Jatropha**
- ✓ Jatropha is a non-food, perennial crop that can be grown on marginal land with a range of climates, soil and water conditions.
 - ✓ It is versatile in a variety of climates, highly resistant to drought, and is able to shed its leaves to conserve water.
 - ✓ There are many countries worldwide that are beginning to invest more in Jatropha.
 - ✓ The largest production is currently in Guatemala which has designated 25,000 acres of land for Jatropha growth.
 - ✓ Additional countries that are investing in this crop include Mexico, the Sudan, Ethiopia, and India.

Table-2: The Feedstocks based on Growth Attributes and Producers

Feedstock (Generation Type)	* Growth Time [47]	* Growth Temperature [47]	* Water Requirements [47]	* Major Growers [48]
Algae (Second)	Dependent on type of algae, temperature and light conditions (authors' assessments)	16–27 °C	Varies with land and sea-based production; Water intensity is generally high; temperature and pH dependent; Light intensity 1000–10,000	Emergent
Corn (First)	110–140 days	18–20 °C	500–800 mm	Brazil, USA, and China
Jatropha (Second)	90 days [49]	16–21 °C [41]	254–1016 mm [41]	Myanmar, India, China, and Indonesia [50]
Lignocellulose (Second)	Varies based on source. Grasses: 3–4 months, waste residue such as corn stover takes as long as the crop from which it is derived [31].	Varies based on source.	Usually need very little water [31]	Emergent
Palm (First)	5–6 months [51]	27–28 °C [44]	Minimum of 1800 mm [44]	Nigeria, Malaysia, and Indonesia
Rapeseed or Canola (First)	85–110 days [52]	Soil temperatures of 3–12 °C [52]	300–600 mm [53]	China, India, and Canada
Rye (Second)	Not available	1–4 °C but below 29 °C for germination [54]	Not available	Germany, Poland, and Russia
Sorghum (Second)	110–140 days	25–35 °C	450–650 mm	Nigeria, India, and Sudan
Soybeans (First)	100–130 days	18–35 °C	450–700 mm	Argentina, Brazil, and USA
Sugar beets (First)	140–200 days	20–25 °C	550–750 mm	France, USA, and Russia
Sugarcane (First)	15–16 months	32–38 °C	1500–2500 mm	Brazil, China, and India
Wheat (First)	100–130 days	15–20 °C	450–650 mm	Russia, China, and India

* The reference within the header applies, unless otherwise noted.

- **Palm**

- ✓ Palm is a prime feedstock for biodiesel, and is produced for biofuels in Indonesia, Malaysia, and other countries of Southeast Asia.
- ✓ Its oil is a fundamental food staple.
- ✓ It is the largest source of vegetable oil consumed worldwide.
- ✓ Palm trees require deep soil, a relatively stable high temperature, and continuous moisture throughout the year.
- ✓ This feedstock grows in rainy and tropical land.

- **Soybeans**

- ✓ Soybeans are a prime fuel and food crop, accounting for 25% and 65% of the global consumption of oil/fats as well as meal/cakes, respectively.
- ✓ The largest producers are the USA and Brazil.
- ✓ This crop can be grown in tropical, subtropical, and temperate climates

- **Sugarcane**

- ✓ Globally, sugarcane is the second largest feedstock for ethanol production, and is a basic food crop that is grown in tropical climates.
- ✓ It can have multiyear harvests tied to a single planting.
- ✓ Sugarcane is grown in deep soil using fertilizers that are high in nitrogen and potassium, and low in phosphorous.
- ✓ Sugarcane requires a constant supply of water throughout the growing season, with varying amounts depending on the climate conditions.
- ✓ Brazil is traditionally the most notable producer of sugarcane-derived ethanol.

- **Sweet Sorghum**

- ✓ Sweet sorghum is a multi-purpose and annual grass crop that is produced mostly by the USA, Nigeria, and India.
- ✓ It is a variety of sorghum that has a high sugar content.
- ✓ It can grow in tropical, sub-tropical, and temperate regions.
- ✓ Relative to sugarcane, sweet sorghum is more versatile, capable of growing with limited water and in poor/shallow soil.
- ✓ Compared to sugarcane and sugar beet alternatives, sweet sorghum is drought-resistant and has a much shorter growing cycle of four months.
- ✓ Given its 70% water content, sorghum must be processed quickly post-harvest.

- **Notable Comparisons**

- ✓ **Table-2** summarizes the major types of feedstock, based on growth attributes and producers.
- ✓ Looking across the feedstock options, Jatropha, rapeseed, soybeans, and wheat have some of the shortest growth periods at 85–130 days.
- ✓ Rapeseed and rye may be consistently cultivated in lower temperatures, whereas sugarcane must be cultivated in a higher temperature environment.
- ✓ Algae, sugarcane and palm have higher requirements for water.

➤ **CLASSIFICATION OF BIOFUELS (FIGURE-8)**

- **Classification According to Food and Agriculture Organization (FAO), USA**

According to this classification, the biofuels are classified into three groups: wood fuels agro fuels and municipal by-products. This classification is according to biofuel sources based on different characteristics, reflected by **Table-3**.

Table-3: Classification of Biofuels by different Characteristics

		woody biomass	herbaceous biomass	biomass from fruits and seeds	others (including mixtures)
		WOODFUELS		AGROFUELS	
Energy crop		- energy forest trees - energy plantation trees	- energy grass - energy whole cereal crops	- energy grain	
By-products*	direct	- thinning by-products - logging by-products	crop production by-products: - straw	- stones, shells, husks	- animal by-products - horticultural by-products - landscape management by-products
	indirect	- wood processing industry by-products - black liquor	- fibre crop processing by-products	- food processing industry by-products	- biosludge - slaughterhouse by-products
End use materials	recovered	- used wood	- used fibre products	- used products of fruits and seeds	MUNICIPAL BY-PRODUCTS
					- kitchen waste - sewage sludge

○ Broad Classification of Biofuels

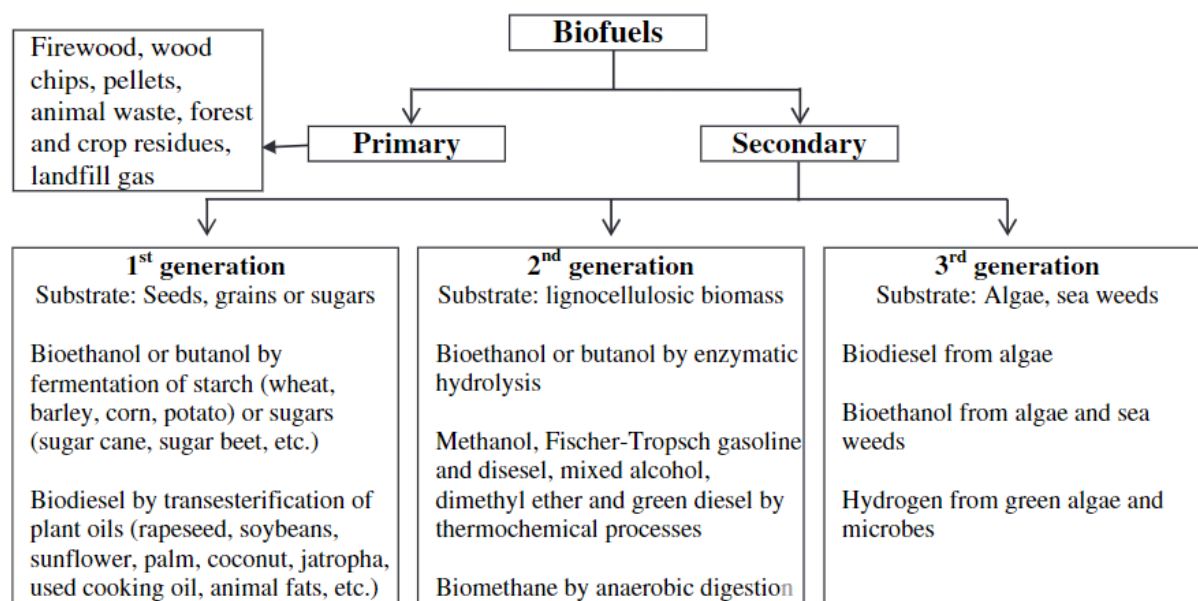


Figure-8: Broad classification of biofuels.

Biofuels are broadly classified as primary and secondary bio-fuels. The primary biofuels are used in an unprocessed form, primarily for heating, cooking or electricity production such as fuel wood, wood chips and pellets, etc.

The secondary biofuels are produced by processing of biomass e.g. ethanol, biodiesel, DME, etc. that can be used in vehicles and various industrial processes.

- ✓ **Primary Biofuels** are natural and unprocessed biomass such as firewood, wood chips and pellets, and are mainly those where the organic material is utilised essentially in its natural and non-modified chemical form. Primary fuels are directly combusted, usually to supply cooking fuel, heating or electricity production needs in small and large-scale industrial applications.
- ✓ **Secondary Biofuels** are modified primary fuels, which have been processed and produced in the form of solids (e.g. charcoal), or liquids (e.g. ethanol, biodiesel and bio-oil), or gases.

The secondary biofuels are further divided into first, second and third-generation biofuels on the basis of raw material and technology used for their production.

○ **Classification based on Generation (Sub-classification of the secondary biofuels)**

According to this classification, the biofuels have been kept in four groups based on the source materials. They are: first generation, second generation, third generation and fourth biofuels.

▪ **First-generation (Figure-9)**

- ✓ The first generation biodiesel is derived from food bio-feedstocks such as soybeans, palm, canola and rapeseed.
- ✓ Promotion of the first generation biodiesel caused interaction problems with human foodchains, including supply and demand balancing, land use, water management.
- ✓ Feedstocks such as sugar, starch, vegetable oil, or animal fats using conventional technology are used.
- ✓ These are generally produced from grains high in sugar or starch fermented into bioethanol; or seeds that which are pressed into vegetable oil used in biodiesel.
- ✓ Common first-generation biofuels include vegetable oils, biodiesel, bioalcohols, biogas, solid biofuels, syngas.

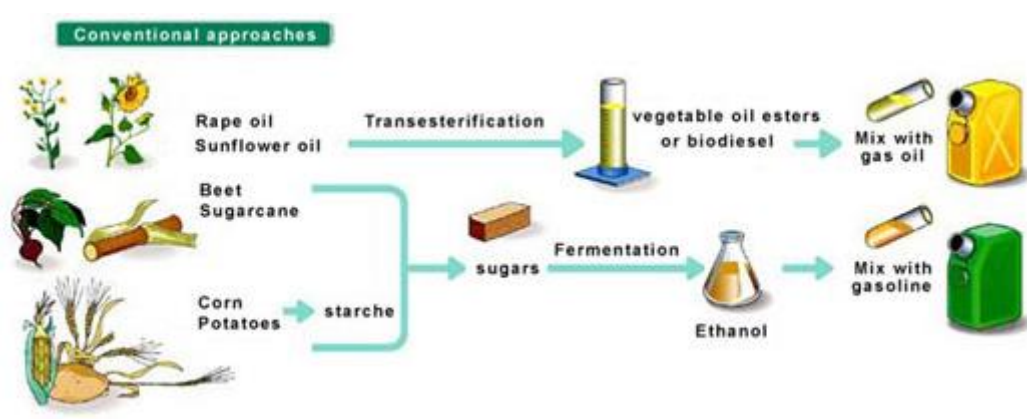


Figure-9: The first generation biofuel.

▪ **Second-generation (Figure-10)**

- ✓ The second generation biofuels are produced from non-food crops, such as cellulosic biofuels and waste biomass (stalks of wheat and corn, and wood).
- ✓ Second generation biofuels are also known as **advanced biofuels**.
- ✓ What separates them from first generation biofuels the fact that feedstock used in producing second generation biofuels are generally not food crops.

- ✓ The only time the food crops can act as second generation biofuels is if they have already fulfilled their food purpose.
- ✓ For instance, waste vegetable oil is a second generation biofuels because it has already been used and is no longer fit for human consumption.
- ✓ Virgin vegetable oil, however, would be a first generation biofuel.
- ✓ Because second generation biofuels are derived from different feed stock, different technology is often used to extract energy from them.
- ✓ This does not mean that second generation biofuels cannot be burned directly as the biomass.
- ✓ In fact, several second generation biofuels, like switchgrass, are cultivated specifically to act as direct biomass.

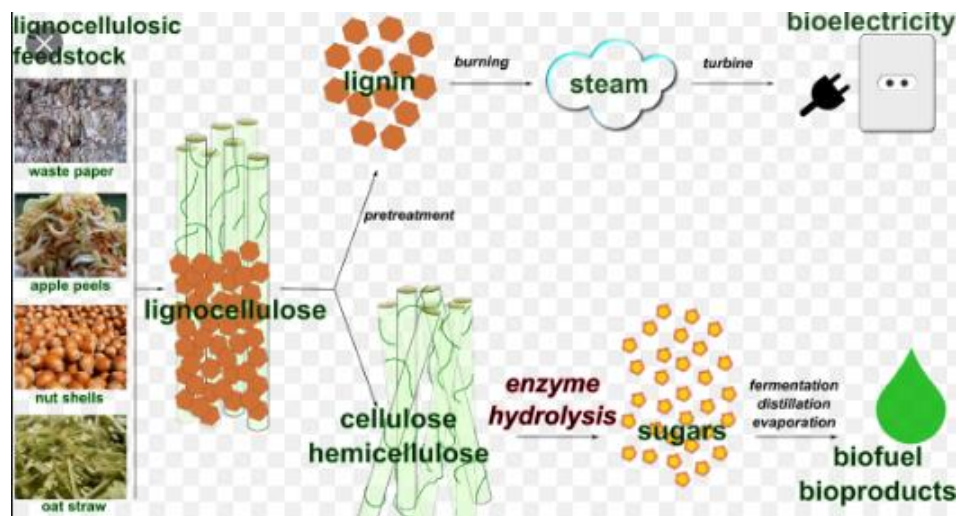


Figure-10: Second generation biofuel.

▪ **Third-generation (Figures-11,12,13)**

- ✓ The term **third generation biofuel** has only recently entered the mainstream it refers to biofuel derived from algae.
- ✓ Previously, algae were lumped in with second generation biofuels.
- ✓ However, when it became apparent that algae are capable of much higher yields with lower resource inputs than other feedstock, many suggested that they be moved to their own category, as result the third category of bioduels was created.
- ✓ Algae provide a number of advantages.
- ✓ The third generation biofuels are – sometimes referred to as “oilgae”.
- ✓ Its production is supposed to be low cost and high-yielding – giving up to nearly 30 times the energy per unit area as can be realized from current, conventional ‘first-generation’ biofuel feedstocks.

Third-Generation Biofuel: Algae

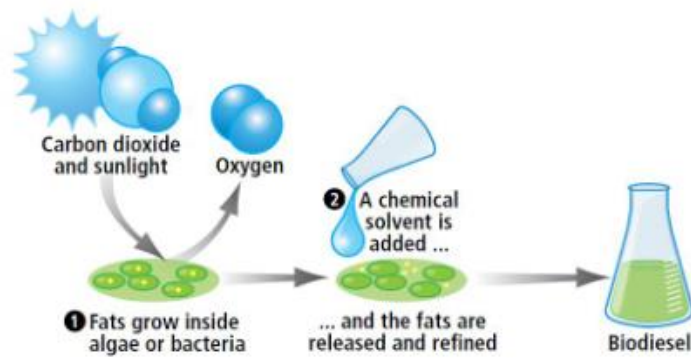


Figure-11: Third generation biofuel.

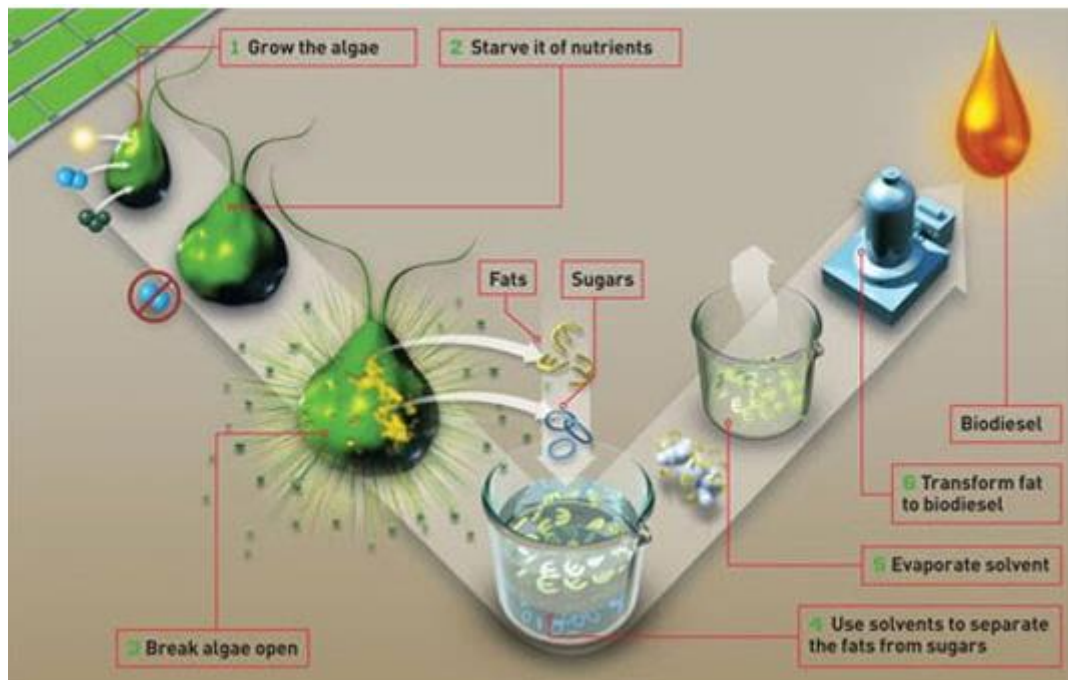


Figure-12: Algae biofuel production.



Figure-13: Algae raceways and photoreactors.

▪ Fourth Generation Biofuel

- ✓ The ecological footprint and economic performance of the current suite of biofuel production methods make them insufficient to displace fossil fuels and reduce their impact on the inventory of Green House Gas (GHG) in the global atmosphere.
- ✓ Engineered algae form the basis for 4th generation biofuel production which can meet this need.
- ✓ The first generation biofuels are known to be made from agricultural products such as corn or sugarcane.
- ✓ The second generation biofuels use all forms of (lingo)cellulosic biomass.
- ✓ The third and fourth generation of biofuel production involves “algae-to-biofuels” technology: the former is basically processing of algae biomass for biofuel production, while the latter is about metabolic engineering of algae for producing biofuels from oxygenic photosynthetic microorganisms.
- ✓ Figures-14, 15 describe the production of fourth generation biofuel.

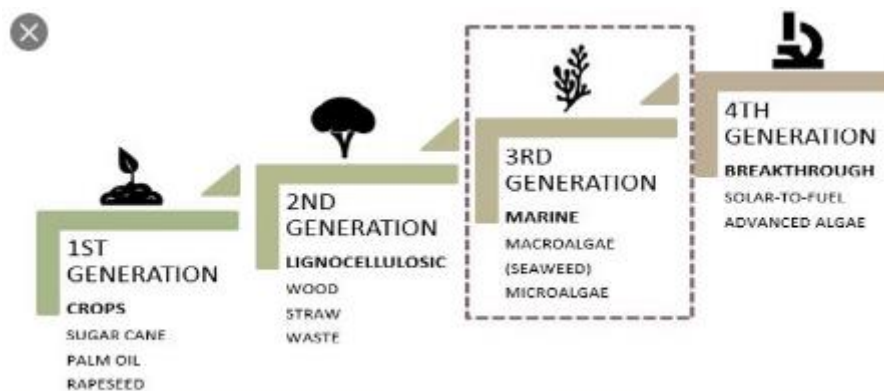


Figure-15: 4th Generation break through from solar to biofuel via advanced algae.

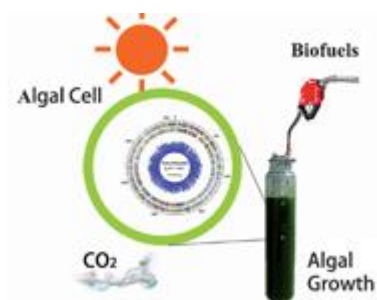


Figure-16: 4th Generation biofuel from engineered algae.

➤ ADVANTAGES AND CHALLENGES OF BIOFUELS

- Biofuels offer the promise of numerous benefits related to energy security, economics, and the environment.
- At the same time, several challenges must be overcome to realize these benefits.
- The main advantages and challenges in the production and consumption of biofuels are highlighted in the **Table- 4**.
- The key advantage of the utilisation of renewable sources for the production of biofuels is the utilization of natural bioresources (that are geographically more evenly distributed

than fossil fuels) and produced bioenergy provides independence and security of energy supply.

Table-4: Potential Benefits and Challenges of Biofuels

Benefits	Challenge
Energy Security Domestic energy source Locally distributed Well connected supply-demand chain Higher reliability	Feed stock Collection network Storage facilities Food-fuel competition
Economic stability Price stability Employment generation Rural development Reduce inter-fuels competition Reduce demand-supply gap Open new industrial dimensions Control on monopoly of fossil rich states	Technology Pretreatment Enzyme production Efficiency improvement Technology cost Production of value added co-products
Environmental gains Better waste utilization Reduce local pollution Reduce GHGs emission from energy consumption Reduction in landfill sites	Policy Land use change Fund for research and development Pilot scale demonstration Commercial scale deployment Policy for biofuels Procurement of subsidies on biofuels production Tax credits on production and utilization of biofuels

- Utilising agricultural residual and waste substrates as raw materials will minimize the potential conflict between food and fuel and also produce the biofertilizer and biopesticides.
- Biofuels produced from lignocellulosic materials generate low net GHG emissions, hence reducing environmental impacts.
- In a report by the United States Department of Agriculture(USDA) the benefits of biodiesel use as fuel included are: it is renewable, suitable replacement for petroleum derived diesel, suitable to use in most diesel engines with no or very little modification, has the potential to reduce GHG emissions, biodegradable with little or no toxicity and can be made from agricultural or other recycled sources.
- Through experiments involving biodiesel produced from different oil types it was found that biodiesel had lower carbon dioxide and polycyclic aromatic hydrocarbons (PAHs) emissions. Biodiesel is considered a “carbon neutral” fuel, as any carbon dioxide released from its burning was previously captured from the atmosphere during the growth of the vegetative crop that was used for the production of biodiesel.
- Biodiesel is said to have a lower flash point than petroleum derived diesel so its transport is safer and easier.
- Besides having several benefits, the production and utilization ZAFRC Dx of biofuels also have several challenges.
- An improved biomass waste collection network and their storage is the main challenge for establishment of commercial biofuel plant.

- A strong policy is needed for organic waste collection and blending of biofuels at higher rate.
- The subsidy for establishment of biofuel plants will accelerate the production of biofuels and tax credits for utilization will create the market for the biofuel.
- The technological improvement could help to improve the system efficiency and provide value added co-products, which will reduce the production cost.

➤ KEY ISSUES AND PERFORMANCE CONSIDERATIONS WITH BIOFUEL SUSTAINABILITY

The following highlights a number of issues and performance considerations for biofuels in the context of sustainability. Implications for ecological systems, society, and the economy are explored.

○ The Food-Fuel Debate

- ✓ The competition for cropland between biofuels and food came to the forefront of public agendas in connection with the volatility of global agriculture prices in 2006–2008, and 2010–2011 [56–60] (Figure-17).
- ✓ From 2006 to 2008, for instance, cereal and oilseed prices doubled in close alignment with overall food index prices.
- ✓ Peak prices for cereal and oilseed increased further in 2011–2012, with corn prices attaining even greater spikes in 2012–2013.
- ✓ Sugar prices tracked more erratically, increasing by more than a factor of 2.5 between 2007 and 2011.
- ✓ It bears noting that price fluctuations are not uncommon, as food price crises occurred in the 1950s and 1970s.
- ✓ However, the degree of volatility and number of affected countries was quite high with the recent surge.
- ✓ For the period between 2006 and 2011, there is broad agreement that the surge was higher than the previous two decades, but lower than what occurred in the 1970s.

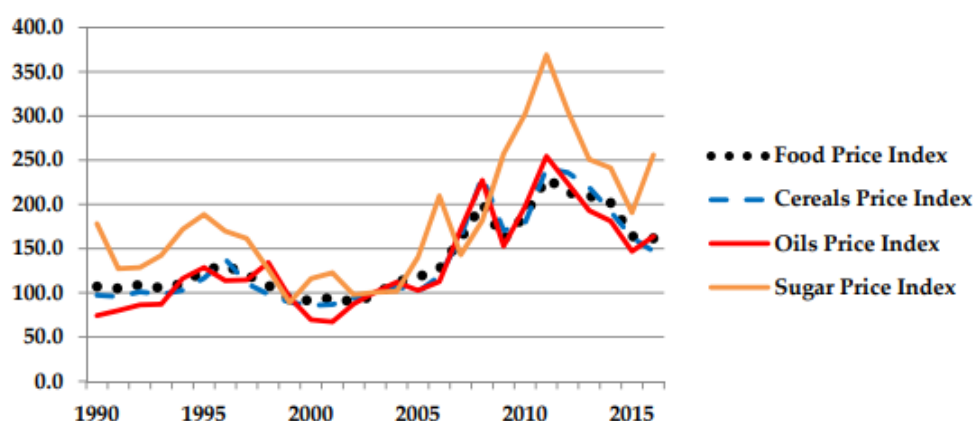


Figure-17: Food price index and key constituents. Data adapted from FAO . The Food Price Index reflects the average of five commodity group price indices.

- ✓ During the period of food price escalation, there were also increases in biofuel production, leading to a concern about the links between food and fuel.

- ✓ In both 2008 and 2011, G20 agendas focused on basic food commodity prices.
- ✓ Concern centered largely on: (1) biofuels impacting food prices, which would disproportionately affect the poor; and (2) land conversion by fuel-based crops that could displace food-based crops or lead to new land appropriation in other areas.
- ✓ No single cause was identified.
- ✓ Analysis indicates that a number of factors were at work, including higher oil prices, weather conditions, investor speculation, and biofuel production.
- ✓ Some estimate that 20%–40% of the rise in food prices was attributable to global biofuel growth.
- ✓ Others point to the complicated mix of factors impacting land use, end-use, and markets for flexible crops.
- ✓ While social, environmental, and economic issues remain to be assessed, debates continue over how to even define and appropriately measure price volatility.
- **Emissions**
 - ✓ Emissions from biofuels became a point of contention in recent years.
 - ✓ This is largely explained by the complex and highly sensitive methods, assumptions, value choices, and localized data differences that were involved, as well as temporal and regional scoping that can affect results.
 - ✓ Not surprisingly, the literature on biofuels reflects a spectrum of competing claims about sustainability.
 - ✓ There are, for instance, claims that the net GHG emissions from biofuels can be worse than those attributed to gasoline in terms of climate effects in the lifecycle assessment scenario.
 - ✓ Alternatively, biofuels are also reported to reduce GHG emissions by 60%–94% relative to fossil fuels.
 - ✓ Looking at regional distinctions, biofuel development in less technologically advantaged countries is also singled out as producing higher GHGs than in more technologically advanced nations.
 - ✓ Here, one can see environmental nuances to biofuel adoption, as well as societal asymmetries, which can in turn produce mixed outcomes in sustainability terms.
 - ✓ Biodiesel, for instance, can favorably reduce particulate matter by nearly 88% relative to petroleum-based diesel.
 - ✓ Yet, this same fuel can also produce mixed results by releasing greater amounts of nitrogen oxides, thereby negatively impacting the environment.
- **Land**
 - ✓ Land use is another primary focus of biofuel sustainability.
 - ✓ As the global population continues to grow, the expansion of land usage can be expected from food, societal expansion, and biofuels.
 - ✓ The United Nations' Food Agriculture Organization estimates that cultivated areas of global land increased by a net 159 million hectares (Mha) or 12% since 1961, during which time irrigated land doubled and agricultural production grew by a factor of 2.5–3.
 - ✓ In this period, concerns over food shortages were met partly by the intensification of fertilizers and pesticides, in addition to new uses of digital information and genetics.

- ✓ A recent estimate indicates that less than 3% of global agricultural land is currently dedicated to cultivating biofuel crops; nonetheless, concerns over land grabbing and indirect land use change for biofuels do exist.
- ✓ Looking ahead, a number of studies on the biomass potential of land shed light on the potential for biofuel production.
- ✓ One calculation found that, of the 13.4 billion hectares (Gha) of the global total land surface, 0.7 Gha is gross available land, with 0.44 Gha reflecting the technical upper limit of what could be utilized for the production of biofuels and other bioenergy by 2050.
- ✓ Here Gha refers to the average productivity of all biologically productive areas (in hectares) on earth in a given year.
- ✓ Of the potential regions for expanding cultivable land, roughly 80% is expected in Africa and South and Central America—primarily in Angola, the Democratic Republic of Congo, Sudan, Argentina, Bolivia, Brazil, and Columbia.
- ✓ In the absence of breakthroughs, such as with agricultural yields, these countries will be pivotal in terms of the environmental, economic, and societal aspects of biofuels, and larger agricultural needs.
- **Water**
 - ✓ Similar to land, water raises questions about the limits associated with biofuels.
 - ✓ To put this into context, roughly 70% of the total world's freshwater is used for agriculture and countries are already experiencing water scarcity issues.
 - ✓ Among the nations affected are 30 developing countries.
 - ✓ By mid-century this number is projected to increase to 55 countries.
 - ✓ Biofuel development could overtax not only the available supply of water, but also the quality of the water, if produced alongside rising food production.
 - ✓ Assuming that agriculture intensifies with more fertilizers to produce fuel crops (to feed a rising population of over 9 billion by 2050), the practice of farming for biofuel feedstock could exacerbate problems in regions that are already challenged by runoff into water aquifers and rivers that create dead zones.
 - ✓ Importantly, there are some approaches to biofuel production that are seen as having a benign effect on water utilization.
 - ✓ Planting switchgrass strategically, for instance, alongside cropland and waterways could minimize nitrate contamination of groundwater.
 - ✓ At this point in time, rising energy demands combined with water resource limitations and hydrologic variability leave a fair amount to be understood in terms of spatial and temporal patterns for water requirements.
 - ✓ It is clear that this remains a key area to monitor for environmental and societal implications.
- **Biodiversity**
 - ✓ Land conversion and its use for biofuels can affect biodiversity, an environmental dimension of sustainability.
 - ✓ Clearing land and forests for cropland may eliminate or disrupt natural habitats for a wide range of species.
 - ✓ However, methods for evaluating biodiversity impacts are still nascent.

- ✓ For lifecycle assessments, the ‘biodiversity damage potential’ and the ‘potentially lost endemic species’ metric are rough gauges.
- ✓ This area has clear relevance for biofuel sustainability determination and related assessments, as with ecosystems services.
- ✓ To date, biodiversity has perhaps been best integrated into biofuel planning with respect to agricultural zoning.
- **Fuel Performance**
 - ✓ If biofuels and fossil fuel counterparts are contrasted, a range of differences is evident.
 - ✓ Favorable octane gains can be observed, for instance, with biofuels, along with less favorable energy fuel economies relative to fossil fuel substitutes.
 - ✓ Higher octane ratings refer to the capacity to withstand compression before igniting.
 - ✓ Specific to ethanol, the fuel economy translates into a reduction of 25%–30% in fuel miles per volumetric unit versus gasoline, whereas for biodiesel, the difference is less (relative to diesel).
 - ✓ If mid-level blends of biofuels are used at 20%–40%, octane benefits can be derived without much energy penalty of the higher blends, though engine and vehicle design changes are required with investment in infrastructure.
 - ✓ Assessing specifically the average emissions impact of biodiesel for heavy-duty highway engines, tests indicate that biodiesel minimizes emissions; however, the decrease depends on the biodiesel source and mixture. If 100% biodiesel is used, combustion produces on average nearly 70% less hydrocarbons, nearly 50% less particulates as well as carbon monoxide emissions, and 10% more NO_x.
 - ✓ Furthermore, the potential for ozone formation by biodiesel is roughly half that of conventional diesel.
 - ✓ Sulfur oxide emissions, an enabler of acid rain, are negligible relative to those from conventional diesel.
 - ✓ With increased biofuel use, one must also factor for a growth in acetaldehyde emissions, which can increase smog and ozone in the atmosphere.
 - ✓ Looking somewhat differently at supply chain emissions for ethanol and biodiesel (excluding land use), ethanol has been estimated at 2–69 kg CO₂-eq/GJ versus 20–49 kg CO₂-eq/GJ for biodiesel, indicating a wider range of environmental impact of ethanol.
 - ✓ Focusing narrowly on a specific fuel type such as ethanol, additional distinctions emerge depending on the feedstock generation type used.
 - ✓ For example, ethanol produced from second-generation cellulosic feedstock requires more energy to break down lignin in cellulose.
 - ✓ However, emissions from the combustion of conventional biofuels can still be greater than in second-generation fuels due to higher overall fuel and fertilizer inputs during production.
 - ✓ Process advances through enzyme efficiency may offer opportunities to reduce the total system-level emissions from second-generation ethanol.

- **Tradeoffs of Fuel Sustainability**

- ✓ With advanced biofuels, greater technical potential exists by drawing upon natural and anthropogenic waste, which could circumvent food-fuel concerns, reduce carbon emissions, and likely reduce pressure on land and water.
- ✓ However, substantial investment will be required, and cost reductions are still needed at the commercial scale.
- ✓ Specific to algae production, the feedstock utilizes less land than other biofuel inputs on a land per volumetric unit of fuel basis.
- ✓ In conjunction with this, fewer impacts can be expected in terms of land, fertilizer, and pest control.
- ✓ In terms of water, currently algae growth has relatively high needs, so algae-based biofuel may not be suitable for water-challenged regions.
- ✓ Technological advances are also needed both to reduce the costs associated with the dewatering step as well as for identifying algal species that produce high yield.
- ✓ If one considers land use with residue-based feedstock for advanced biofuels versus that required for grain and dedicated crops in conventional biofuels, the former is more favorable since no additional land is required.
- ✓ Such a scenario avoids competition for land, and, in turn, has minimal impacts on food prices as well as GHG impacts and likely water.
- ✓ Crop residue removal for advanced biofuels could also have positive impacts on pest and disease control.
- ✓ Yet residue utilization can also be disadvantageous, since crop residues also conserve soil properties, enhance soil productivity, sequester carbon in soil, and conserve water.
- ✓ Such tradeoffs merit further investigation.
- ✓ **Table-5** presents data on a number of biofuels' attributes, including water and energy.
- ✓ As with life cycle assessments, one must factor for variations in scoping and accounting.
- ✓ Looking across feedstock options, water needs are highest for algae, rapeseed, and sugarcane.
- ✓ The energy balance is highest for rapeseed-based fuel and cellulosic ethanol.
- ✓ Somewhat differently, the energy intensity reflects biodiesel at much higher values than ethanol per volumetric unit.
- ✓ If yields as well as fertilizer requirements are factored in, some estimates indicate that the energy return on investment for sugarcane ethanol relative to corn-based ethanol is 4–6 times larger.
- ✓ System disturbances represent another area of consideration.
- ✓ Broadly speaking, accidents with biofuels may be less dangerous to the environment relative to fossil fuel substitutes, since biofuels will more readily biodegrade.
- ✓ If indirect impacts are considered, such as those associated with the use of tallow and cooking oil in biofuels, there are benign effects in the removal of such waste products from the system that could contaminate ground water.

Table-5: Comparison of Biofuel Sustainability Characteristics

Biofuel	GHG Emissions CO ₂ e/MJ	Water Intensity L/L Product	Energy Intensity MJ/L	Net Energy Balance MJ/L Product
Gasoline (Baseline)	94 g *	2.8–4.6 *	35.4 [90]	28.3 *
Corn Ethanol	76 g: major contributors 31 g (ethanol production) and 17 g (fertilizer) [90]	175.4 [91]	21.3 [90]	10.1 [90]
Sugarcane ethanol	45 g (includes 16 g from land use change) [90]	526 [91]	21.3 [90]	16.4 [90]
Soybean biodiesel	59.19 g [92]	369.2 [91]	32.7 [92]	
Rapeseed/canola-based biofuel (Biodiesel)	59.19 g [92]	645.5 [91]	32.7 [92]	21.6 [93]
Cellulosic ethanol	43 g [94]	6.5 (Switchgrass) 387 (drought conditions) [95]	21.3 [90]	21 (Switchgrass) 20.4 (corn stover) 21.4 (miscanthus) [90]
Algae biodiesel		44 (enclosed production) 216 (open production) [95]	32.7 [92]	

Note: (1) Energy intensity gauges the amount of energy released from combustion of a fuel, in this case, measured in MJ released per liter of biofuel; (2) Water intensity is the amount of water required to produce a fuel, here measured in liters of water needed per liter of fuel produced—this includes the water needed in the production of the biofuels and the water needed to grow the feedstock; (3) Energy balance is the net energy in the product after deducting the total required energy to produce the fuel. * Calculated.

➤ **POLICY CONSIDERATIONS**

Policy has played a pivotal role for biofuels and will likely continue for the foreseeable future by encouraging or impeding sustainable approaches, reducing barriers, and highlighting information or funding needs. This section outlines policy approaches for key biofuel-producing regions. A critical review follows, outlining unsustainable system issues with policy-related trade activity.

• **Brazil**

- ✓ In Brazil, the blending requirement for ethanol recently has been 18%–27.5%, currently 27%.
- ✓ Rules for the biodiesel mix designate a stepped timetable to increase the mix from 7% to 10% by 2019.
- ✓ A regional producer subsidy for ethanol is in place to more evenly balance costs of production between less and more developed growth regions.
- ✓ In conjunction with the economic downturn, no support was provided in 2015.
- ✓ Tax incentives exist for ethanol-conductive vehicles, which translate as a reduced tax burden for flex fuel vehicles versus that for strictly gasoline-only fueled vehicles.
- ✓ Specific to biodiesel, the National Biodiesel Production Program (PNPB) was launched in 2004, compelling suppliers to procure vegetable oil from small producers and family farms.
- ✓ Tax exemptions and incentives are in place for biodiesel, based on feedstock, producer size, and region, in order to encourage production and social inclusion.
- ✓ An intricate tax policy system also exists for fuels, spanning local to federal jurisdictions that are currently set to be favorable for ethanol versus gasoline.

- ✓ However, the setting of artificially low gasoline prices to counteract inflation in recent years had a deleterious effect on biofuels.
- ✓ Support for project financing in the form of investment credit lines is also indicated, yet it is unclear how readily these funds will be available given the current economic conditions.
- **China**
 - ✓ Bioenergy is a dimension of China's strategic energy planning.
 - ✓ Biofuel programs have been implemented since the early 2000s, with direct subsidies for conventional grain-based biofuels now discontinued.
 - ✓ The 12th Five Year Plan that ended in 2015 included a goal of producing 4 million tons of fuel ethanol and 1 million tons of biodiesel.
 - ✓ Broadly, the country has a 15% biofuels target by 2020 and aims to move toward a 10% mandate.
 - ✓ In addition, a number of the provinces have mandates in place to blend 10% biofuels.
 - ✓ With China taking an active role in curbing CO₂ emissions, more policies to encourage renewables including biofuels are anticipated.
- **European Union (EU)**
 - ✓ Regulations for the use of transport-based biofuels are outlined in the 2009 EU Energy and Climate Change Package (CCP).
 - ✓ The CCP includes requirements which stipulate that 20% of the overall EU energy mix should be renewable energy in 2020.
 - ✓ Within the CCP, the Renewable Energy Directive (RED) defines sustainability requirements for liquid biofuels, encompassing GHG reductions, land management, as well as additional environmental, social and economic criteria.
 - ✓ In 2015, the European Commission established a 7% cap (energy basis) on conventional, food-based biofuels in transportation by 2020, which limits future production of Generation 1 biofuels.
 - ✓ It resides within a larger 10% target in the RED that is obligatory for all member states.
 - ✓ Advanced (non-food) biofuels are noted with a non-binding five percent national target.
 - ✓ Member states have until 2017 to implement the revised rules.
- **India**
 - ✓ In 2009, a national biofuels policy was instituted, encouraging the use of renewable energy in transport, with an aim to replace 20% of petroleum-based fuel with biofuels by the end of the 12th Five Year Plan in 2017.
 - ✓ In 2014, diesel prices were deregulated, enabling more favorable conditions for biodiesel production.
 - ✓ The government announced a blending requirement of 10% ethanol in gasoline, beginning with the October 2015/2016 sugarcane season, alongside existing rules that set minimum sugarcane pricing.
 - ✓ Discussions are now focused on amending the 2009 biofuels law, including coverage of a mandatory blend for biodiesel.

- ✓ A recent push by India to become a “Methanol Economy” and a net zero petroleum import country is an endeavor to watch in the coming years with respect to the impact on biofuels and national policy associated with alternate fuels .
- **United States**
 - ✓ The USA requires the use of a minimum volume of biofuel in transportation, but does not mandate its production.
 - ✓ This policy is enshrined in the Renewable Fuel Standard (RFS) that was established with the Energy Policy Act of 2005 and later enlarged with the Energy Independence and Security Act of 2007.
 - ✓ The Environmental Protection Agency oversees the RFS mandate, which essentially is designed to increase consumption volumes from 9 billion gallons of renewable fuel in 2008 to 36 billion in 2022.
 - ✓ The RFS outlines four categories of fuels that meet the statutory requirements, with sub-mandates existing for various advanced fuels.
 - ✓ The EPA regulates compliance with a tradeable credit system, and waiver capabilities.
 - ✓ The agency is required to announce volumetric requirements each November for the upcoming year, with the exception of biomass-based diesel, which must be announced 14 months in advance.
 - ✓ Tax credits for blending provide \$1.00 per gallon of biodiesel, agri-biodiesel, or renewable diesel that is blended with petroleum diesel to produce a mixture that includes at least 0.1% diesel fuel.
 - ✓ Related tax credits exist for delivery of 100% biodiesel as an on-road fuel.
 - ✓ Feedstock incentives provide financial support to establish biomass feedstock crops for advanced biofuels facilities and produce advanced biofuels.

➤ **INVESTMENT IN BIOFUELS**

Looking beyond the technical aspects, sustainability/performance, and policy aspects of biofuels, another critical dimension of the biofuel outlook can be found in investment trends.

- Global investment in biofuels was estimated to equal \$3.1 billion in 2015, reflecting a decline of 35% relative to 2014 and more than 80% in nominal terms since 2008.
- At the beginning of the 21st century, billions of dollars were invested in advanced biofuel projects by international oil companies, with many of the projects later being abandoned.
- Commercialization of advanced biofuels is more costly and protracted than originally anticipated.
- The decline in per barrel oil prices from \$115 in June 2014 to \$27 in 2015 recovered somewhat to roughly \$50 in most of 2016, yet the absence of a coherent biofuel policy in the United States for much of 2015, plus necessary time and financing, have combined to deter all but a small set of investors.
- Key players that are crucial for enabling investment and other development at scale for biofuels will continue to include airlines and other, large corporations.
- Among airlines, United Airlines signed a \$30 million deal in 2015 with Fulcrum Bioenergy to provide alternative jet fuel at prices that are competitive with conventional jet fuel.

- JetBlue also signed a 10-year power purchase agreement with S.G. Preston for renewable jet fuel that is produced from non-food, hydro-processed esters and fatty-acid-based feedstock.
- Outside of aviation, large companies like Dupont (known for its historical strength in chemicals and ammunition) are driving the development of cellulosic plants.
- Global production capacity for advanced biofuels at the end of 2015 was estimated to be 225 million gallons per year.
- Planned capacity would add another 390 million gallons per year, with initiatives underway in Brazil, China, Canada, the Netherlands, the United Kingdom, Sweden, France, and the USA.
- Notably, the majority of the existing capacity is in ethanol as shown in **Figure-18**.

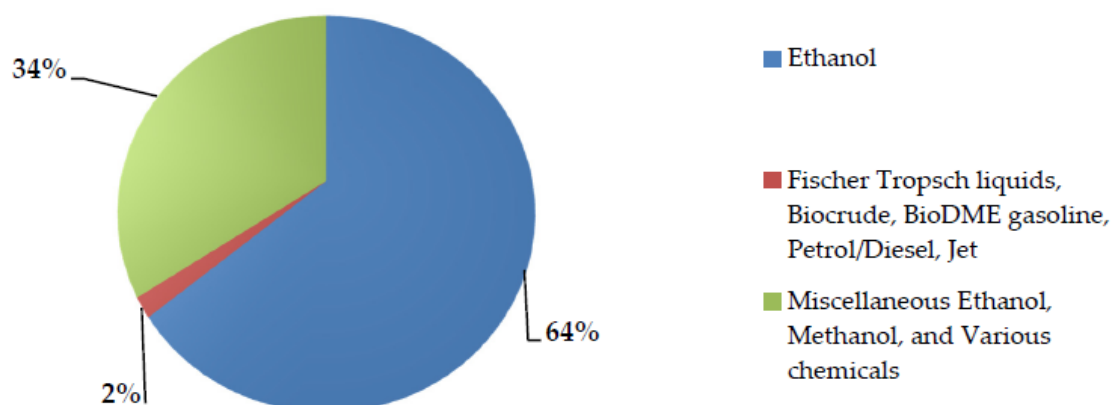


Figure-18: Advanced biofuels by fuel products—operational, 225.43 million gallons/year.

(Note-Advanced Biofuel: Also called second generation biofuels, any biofuel produced from a sustainable feedstock that does not threaten the food supply).

- When viewed from the standpoint of development stages for advanced biofuels, more than 80% of existing capacity is commercialized as evident from **Figure-19**, the majority of which started in 2014 or later.
- In 2015 and early 2016, two commercial-scale, advanced biofuel plants were commissioned (Finland and the United States), plus three pilot-scale demonstration plant.
- The performance of these plants could be pivotal for future development.

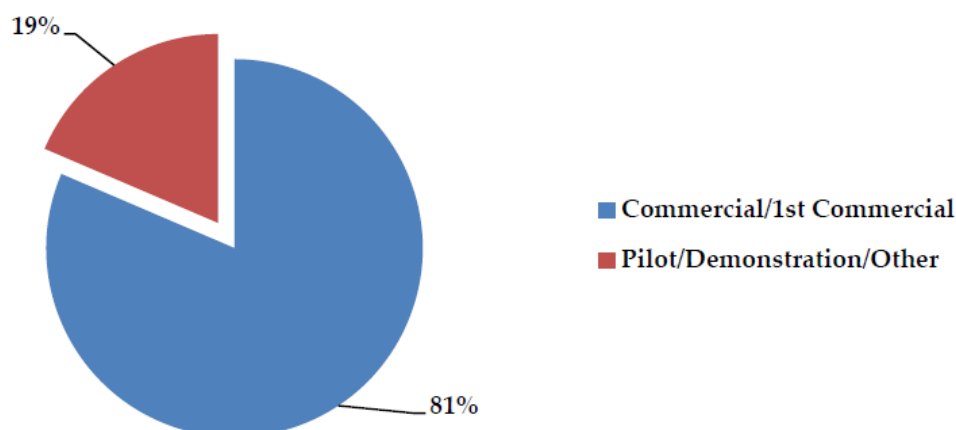


Figure-19: Advanced biofuels capacity per year—operational, 225.43 million gallons/year.

- Important, continuing challenges remain for advanced biofuels in ensuring sustainability and reducing production costs.
- The most challenging aspect of algae-derived fuels, for example, is the dewatering step.
- Specific to production costs for advanced biofuels, recent estimates are significantly higher than \$3 per gallon, which substantially exceeds the untaxed price of petroleum alternatives.
- One can look, for instance, at the Abengoa SA, owner of a cellulosic biofuel plant in Kansas, which suspended operation in late 2015 and filed for bankruptcy.
- The plant was built with loan guarantees of \$132.4 million and a \$97 million grant from the U.S. Department of Energy.
- This example underscores not only the challenges of developing economically competitive, advanced biofuels, but of managing technological progress in an emerging market.